Models and AI Methods

Task	Technology	Role
Ingestion	PyMuPDF	Extract text, blocks, pages
Cleaning / chunking	Python	Normalize PDF text and section chunks
Embedding	all-MiniLM-L6-v2 via Sentence Transformers	Pretrained text-to-vector inference
Retrieval	ChromaDB	Similarity-ranked evidence search
Answer generation	Google Gemini	Grounded natural-language output
Test provider	MockLLMProvider	Repeatable offline responses, no API costs
Citation verification	Python CitationVerifier	Validate retrieved citation IDs
Entity extraction	Regex/table rules, optional LLM fallback	SWC/Port/Interface records

No new transformer was trained or fine-tuned. No PCA/t-SNE/UMAP dimensionality reduction is implemented. A sentence embedding is a learned vector representation, not PCA.

Model ID discrepancy: config/settings.py (../config/settings.py) defaults to gemini-3-flash-preview, whereas root .env.example (../.env.example) specifies gemini-2.5-flash. The real effective model depends on the private .env used for the specific run; do not report a model that was not verified.

Responsible interpretation: a valid citation identifier confirms source provenance, not that each claim is factually supported. A human reviewer checks the quoted text.

Configuration and Retrieval Parameters

Source of truth: config/settings.py (../config/settings.py), coordinator (../backend/coordinator.py), retriever (../core/retrieval/retriever.py), prompt builder (../core/rag/prompt_builder.py), evaluator (../core/evaluation/evaluator.py).

Parameter	Actual implementation/default
Embedding model	all-MiniLM-L6-v2
Embedding device	cpu
Gemini model	gemini-3-flash-preview Settings default vs gemini-2.5-flash in .env.example
LLM default	mock, can select gemini
Vector path	./data/chroma_db
SQLite path	./data/autosar_assistant.db
Retriever top_k	5 default
Retriever similarity threshold	None default (no filter until explicitly passed)
Coordinator query threshold	None default
Golden evaluator threshold	0.3 function default; application differs
Prompt context	12,000 characters max by default (approximate token budget)
Q&A temperature	0.0 supplied by coordinator
API / UI	127.0.0.1:8000 / localhost:8501

Distinctions: Top-k is a candidate count, not relevance confidence. Similarity is not a calibrated probability of correctness. The evaluator's keyword-match ratio is not a semantic correctness score. In core/evaluation/evaluator.py, the relevance-threshold argument is treated differently for answerable/unanswerable items; inspect its implementation before quoting system-wide results.

Do not upload actual .env or runtime credentials. Verify the model and retrieval settings displayed during each real demonstration.

Grounded RAG Prompt: Actual Structure

Exact authoritative implementation: core/rag/prompt_builder.py (../core/rag/prompt_builder.py).

• Trusted system message contains AUTOSAR engineering rules and prohibits inventing software components, port relationships, signals, runnables and other facts.

• Untrusted PDF content is enclosed in <context> and per-evidence <chunk id="..." doc="..." page="..." section="..."> elements.

• User question is isolated in <user_question>.

• All factual claims must cite the exact retrieved [chunk_id].

• Missing evidence requires: Based on the provided AUTOSAR specification, there is insufficient evidence to answer this question.

• The builder escapes certain XML-like delimiter sequences, reducing simple document-context breakout attacks (not a complete injection defense).

• Context accumulation is capped at 12,000 characters by default.

Illustrative interaction:

Question: “What does DoorControlSWC do?”

Relevant source: “DoorControlSWC processes lock/unlock requests and publishes door status.”

Desired generated output: “DoorControlSWC processes lock and unlock requests and publishes door status [ACTUAL_RETRIEVED_CHUNK_ID].”

ACTUAL_RETRIEVED_CHUNK_ID is a placeholder, not a valid live citation. Never fabricate IDs in an evaluation.

Real limitation: MockLLM has previously generated generic citation-bearing answers that were not fully responsive and failed to refuse an unsupported MCU question. Semantic claim support must be checked manually; this behavior is not a passed hallucination test.

Token management: reuse local embedding model, default top-k 5 and MockLLM for deterministic tests; send small retrieved context to Gemini only when needed.